

CS 2100 students will complete an open-ended semester-long project to apply the skills introduced in the course. There will be five checkpoints during the semester. Each checkpoint will have topics that each student must incorporate into their semester project, but it is ultimately up to the student to incorporate the required topics and build something unique.

By the **checkpoint** due date, students must incorporate the specified topics into their project and submit it to Pawtograder. Then, after that, during the associated **check-in meeting** (at a time assigned to each student during lab), each student will discuss with a TA 1) how they incorporated the topics, and 2) how they plan to incorporate the next topics into their project before the next checkpoint.

Specifically, these are the checkpoints with the associated topics that students will incorporate into their semester projects:

	By this date:	Students must incorporate this into their project and submit it to Pawtograder:
1	October 1	Choose a dataset for analysis that interests you. This dataset must represent something in the real world. Example datasets include: • This dataset of real and fake news • This dataset of infectious diseases in California • This dataset of public data requests in Oakland • This dataset of 311 calls in San Francisco • This dataset of 311 calls in Boston An easy way to find data is to go to Kaggle and search for something that interests you. The dataset must have at least three numerical variables and at least one categorical variable, and it must have at least 100 rows. You may process the data to calculate numerical or categorical variables. (E.g., you may count the number of times a word appears in some text, or you may create a boolean variable for whether the text is more than 50 characters long.) You may also select a sample of the dataset, if it is a big dataset. Discuss: • How was the data collected? What are some implications of the way it was collected? (E.g., is there selection bias?) • What applications can this data be used for? Good or bad?
2	October 15	• Design and implement a class to represent something in the dataset. • Create a stakeholder-value matrix with at least five stakeholders and at least five values, to discuss an application of the dataset.
3	October 29	Write Python code that performs data analysis of the dataset:

		• Independent analysis of each numerical and categorical variable, e.g., mean, variance, min, max, distribution, rate of missing values • How are the variables related? Are you able to discern any patterns or associations? • What are the real-world interpretations of your analysis?
4	November 12	Create a plan including a UML diagram for an object-oriented project using the dataset that you selected (or, if necessary, a new dataset). It must have: • At least three classes which are related via subclass - superclass relationships • At least two properties which are not attributes • A main function that uses all of the classes involved and has some output (printed to the terminal, output to a text file, audio or video files, a webpage, etc.) For example, you may create a program that calculates the winner of a sports league, sorts 311 calls by priority, or filters spam out of email.
5	December 2	Update on implementing the project. Topics to incorporate: • A class that implements the comparison protocol • A class that implements the Iterable interface (can be the same class or a different one) • At least two abstract classes or interfaces • At least two static or class methods (or one of each)

It is recommended that students use this project to:

- Learn more about each checkpoint's associated topic
- Practice the skills introduced during class time and other assignments
- Build something unique that they can show off (e.g., on a resume)
- Learn about other Python features and libraries which are not covered in class, and use them in an interesting way!

Students will receive the grade for each checkpoint during the scheduled meeting with the TA during lab. To receive full credit for each checkpoint, the student must show how they have successfully incorporated the topic into the project (by discussing code submitted via Pawtograder).